

Motivation

In 1980, Vera Rubin published a paper that is credited with providing the first substantial evidence for dark matter (Rubin, Ford, and Thonnard 1980). However, no one has detected dark matter particles thus far. Rubin's paper focuses primarily on the issue of galactic rotation curves.

A further mystery the lopsidedness of galaxy spectra, where the kinematic center does not overlap with the luminosity center of more than 35% of the galaxies observed in (Jog and Combes 2009; Garcia-Lorenzo et al. n.d.).

Hypothesis

All particles share a universal spatial mass profile:

$$m = f \phi(\mathbf{x}),$$

where f is particle-species-specific and $\phi(\mathbf{x})$ is a universal function of position. Near Earth, $\phi \approx \text{const}$, so local experiments cannot falsify the hypothesis. On galactic scales, deviations in ϕ become observable.

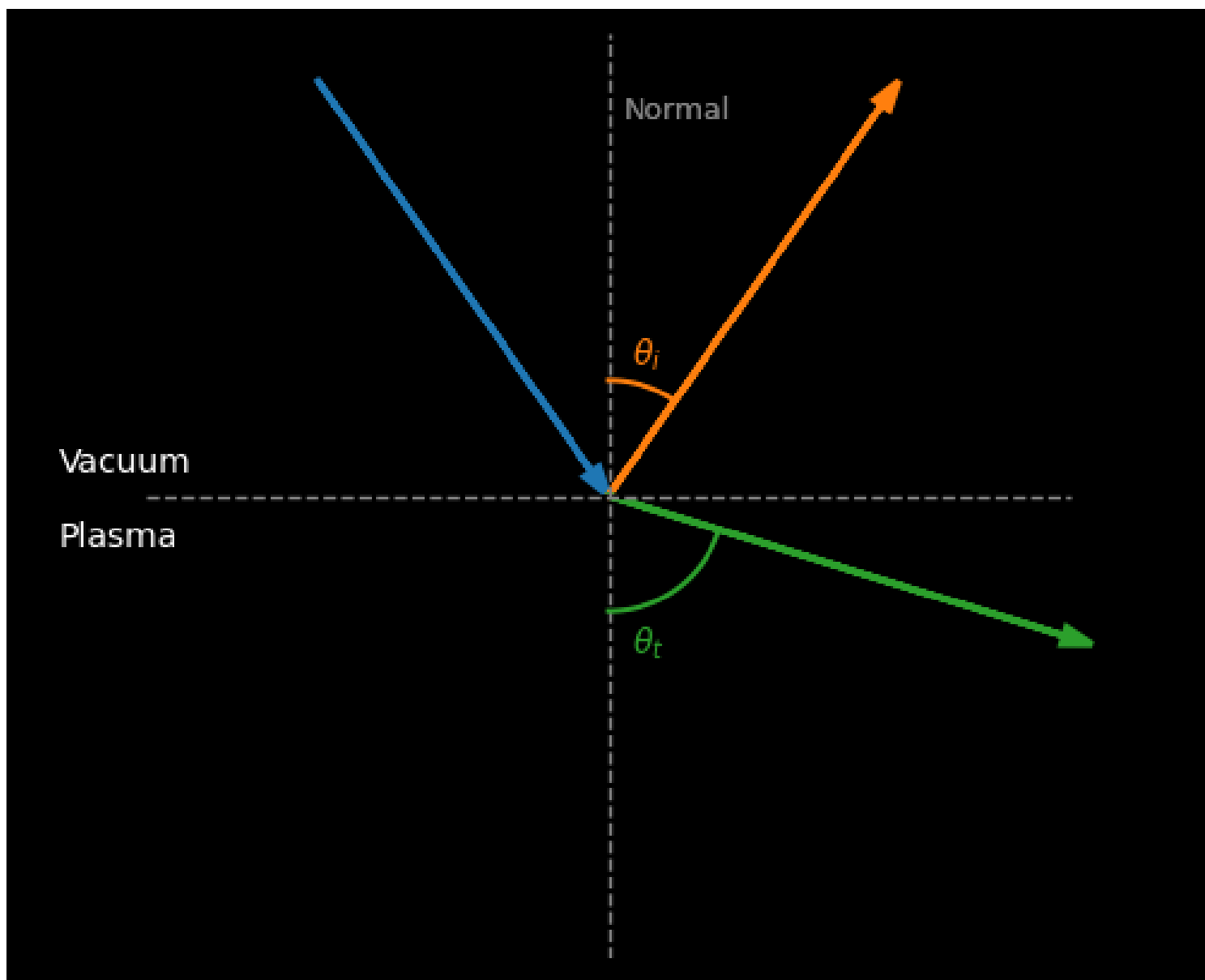
Plasma Analogy

There is an analogue to the mechanics of variable mass particles that comes from plasma physics. Photons in a plasma satisfy

$$\omega^2 = \omega_p^2 + k^2,$$

which mirrors Einstein's energy-momentum relation $E^2 = m^2 + p^2$. Identifying the photon mass with ω_p , which depends on local plasma density, shows that a spatially varying density acts as a spatially varying mass. Density gradients cause photon trajectories to curve, interpretable as a mass-gradient effect.

We generalise: if photons acquire an effective space-dependent mass via the medium, other massive particles may be subject to an analogous mechanism on cosmic scales.



Key Insight: A space-dependent mass can produce trajectory deviations that mimic the gravitational effects attributed to dark matter, without requiring new particles.

Methods

One-Dimensional Case

To build the method, we start from a modified relativistic Lagrangian with a variable mass:

$$\mathcal{L} = -m(z)\sqrt{1 - \dot{z}^2}.$$

This is paired with the constant-proper-acceleration trajectory

$$z(t) = \frac{1}{a} \left(\sqrt{a^2 t^2 + 1} - 1 \right).$$

Using the Euler-Lagrange equations and the known trajectory, we invert the problem to recover $m(z)$. The method is validated because it exactly recovers the original (constant) mass. A Lipschitz condition guarantees **uniqueness** at every point in space and time, with an expected singularity at the Rindler horizon.

Two-Dimensional Orbital Case

For orbital mechanics, the modified central-force Lagrangian is

$$\mathcal{L} = \frac{\mu(r) \dot{r}^2}{2} + \frac{\mu(r) r^2 \dot{\phi}^2}{2} + \frac{GmM}{r},$$

where $\mu(r)$ is the position-dependent reduced mass. The Euler-Lagrange equations reduce to an ODE for $\mu(r)$ given an observed trajectory. It is unknown wether or not an analytic solution exists and the dynamics should be aperent from numerical methods, so we employ a guess and check method to mass profiles by integrating the equations of motion with a **4th-order Runge-Kutta** scheme.

The effective potentials for the three cases take the form:

$$V_{\text{eff, classical}} = -\frac{GM}{r} + \frac{L^2}{2\mu r^2}$$

$$V_{\text{eff, relativistic}} = -\frac{GM}{r} + \frac{L^2}{2\mu r^2} - \frac{GML^2}{\mu r^3}$$

$$V_{\text{eff, spatial, classical}} = \frac{\mu(r) \dot{r}^2}{2} - \frac{GM\mu(r)}{r^2} + \frac{L^2}{2\mu(r)r^2}$$

The RK4 integrator has been tested against known newtonian and relativistic analytic solutions.

Methods Continued

The next step is to extremize the swartzchild solution with the varying mass and find V_{eff} for the variable mass relativistic case.

$$V_{\text{eff, spatial, relativistic}} = ?$$

Results & Progress

- 1D case complete.** Recovered the expected Lagrangian from an observed trajectory; uniqueness proved via Lipschitz condition.
- 2D non-relativistic potential derived.** Found the modified V_{eff} for the spatial-mass case and implemented RK4 simulation validated on classical and relativistic orbits.
- Analytic two-dimensional newtonian solution under investigation

Timeline

Period	Milestone
Nov – Feb	Reproduced 1D case; uniqueness proof
Mar – Apr	2D non-relativistic modified potential
May	Mass distribution form + matching trajectory
Jun – Jul	Analytic or numerical form; compare to observational data
Aug	Begin thesis writing

References

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